

Illustrative render of the finished design - not a photograph of a sample

Bobble Snowflake Tree Skirt

Twelve-spoke double-crochet circle · closed centre · three sizes · exact scalloped border

CROCHET PATTERN · US + UK TERMS SIDE BY SIDE · INK-FRIENDLY PRINT LAYOUT

✓ 5-AXIOM MATH VERIFIED ON PAPER | **32 ROUNDS** | **3 SIZES · 46-109 CM** | **5-5.5 MM HOOK** | **WORSTED #4**

● FOREST GREEN - MC ○ OAT CREAM - CC named original colourway

WHAT YOU GET	VERIFICATION	PRINT NOTES
32 numbered rounds, US + UK columns Printable tick-box count checklist Spokes, scalloped border, three sizes	✓ 5-axiom mathematical audit 32 / 32 rounds and closures pass Paper-verified: no sample crocheted	10 pages on US Letter, light ink load No full-bleed colour pages Tables sized for home printers

One verified ladder, three stopping points

mini 46-53 cm · standard 74-84 cm · large 97-109 cm - each ends on a count divisible by six, so the exact scalloped border closes with 28, 46 or 64 scallops.

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Personal & small-batch finished-item licence · Design Code NS-14 · pattern text machine-verified (crochet-check v1.0.0)
Cover artwork is an illustrative render of the written design - not a photograph of a sample.

Welcome, Contents & How to Use This Pattern

A closed-centre, twelve-spoke double-crochet circle with a contrast bobble round every third round from R5. Choose the mini, standard or large stopping point, add twelve optional surface-crochet spokes, then finish with an exact scalloped border.

Tick each **[ ]** box only after the row's stated count matches your work. Work from one column only: US on the left, UK on the right. If a count differs, stop and recount the round before continuing. Diagrams support orientation; the numbered written directions control.

Welcome, Contents & How to Use This Pattern 2
Safety & Materials 2
Gauge, Size & Abbreviations 3
Construction & Techniques 3
Instructions - Centre Ring & Growth Rounds 4
Spokes, Scalloped Border & Sizes 6
Finishing, Assembly & Troubleshooting 7
Colourways & Care 7
Terms, Licence & Finish Checklist 8
Appendix A - Visual Schematics, Assembly Map & Growth Chart 9
Appendix B - Machine-Verification & 5-Axiom Audit 10

Format	US + UK terms side by side	**Skill**	Easy-intermediate
Sizes (target, verify)	Mini 46-53 cm / Std 74-84 cm / Large 97-109 cm	**Yarn**	Worsted #4: forest green MC, oat cream CC
Hook	5-5.5 mm (US H/8-I/9)	**Construction**	Closed joined rounds, no turning

Safety & Materials

Home decor, not a toy - and not flameproof
Keep it away from candles, fireplaces, heaters, hot lamps and every ignition source; use only cool-running lights approved for the tree. Do not cover plugs, adapters, power strips or electrical connections, and route cords so the skirt cannot pull on them. Keep the edge clear of walking routes; it must not support, level or stabilize a stand. Fit the closed centre without obstructing the stand, reservoir or fasteners. No finished NS 14 skirt has been independently crocheted, fit-tested, wash-tested or assessed for flammability; before sale or supply the maker or seller must determine every applicable assessment, testing, documentation, labelling and traceability duty.

- Worsted or aran (#4) yarn. Original colourway: forest green main colour (MC) and oat cream contrast colour (CC), in comparable yarns that meet the same gauge.

Planning quantities - estimate, not a weighed result
Planning quantities below are estimates derived from the written stitch counts, not weighed results. The source planning allowance of 700–1,200 g was never measured, is far above what the stitch counts imply, and supplied neither a mini figure nor a colour split. Estimated total yarn at about 200 yd / 100 g of worsted and 3–4 in of yarn per US dc (a 5-dc bobble uses five dc of yarn but counts as one stitch, and the estimate includes the spokes and border): **mini 70–95 g · standard 165–225 g · large 305–410 g**. Because every bobble round is worked entirely in CC and CC also covers the spokes and border, plan a near-even split: **CC about 44–52 % of the total, MC the remainder**, with the CC share highest in the mini. Make and weigh a gauge piece and a complete sample before buying for production or publishing a quantity, and buy both colours from one dye lot.

- 5–5.5 mm crochet hook (US H/8–I/9), or the size needed to meet the target stitch gauge while keeping the circle flat.
- Optional hook 0.5–1 mm smaller for loose surface slip stitches; do not use it if it contracts the fabric.
- Tapestry needle, scissors, tape measure, digital scale, one start-of-round marker and, for the optional spoke route, twelve locking markers or removable guide threads.
- Rust-resistant pins and a fibre-appropriate blocking surface large enough for the selected size.

Size	Stop	Growth sts	Est. yarn	CC share
Mini / tabletop	R14	1,260	70-95 g	~52 %
Standard	R23	3,312	165-225 g	~47 %
Large	R32	6,336	305-410 g	~44 %

Gauge, Size & Abbreviations

Make the centre ring and work through R6 as a circular gauge piece; let it rest before measuring. Target gauge in US double crochet / UK treble crochet is **12 stitches = 4 in / 10 cm** across the stitch direction.

Radial gauge is derived, not independent
The radial figure is derived from that stitch gauge, not independent of it: each round adds 12 stitches, so the circumference grows 4 in / 10 cm per round and the radius of a flat circle must grow 4 / 2π = **0.64 in / 1.6 cm per round — about 6¼ completed rounds = 4 in / 10 cm radially**. A rounded "6 rounds = 4 in" makes every round roughly 5 % taller than flat geometry wants, so the circle carries a mild cupping tendency that blocking removes; do not treat a small cup as a count error. Match the stitch gauge first, then confirm the circle lies flat. Stitch gauge controls circumference; the increase rate controls flatness. A rectangular swatch alone does not verify how this twelve-increase circle will lie.

At the target stitch gauge, the unbordered circumference and derived diameter are approximately:

- **R14** 168 stitches / 3 stitches per inch = 56 in circumference; 56 / π = 17.8 in / 45 cm diameter.
- **R23** 276 / 3 = 92 in circumference; 92 / π = 29.3 in / 74 cm diameter.
- **R32** 384 / 3 = 128 in circumference; 128 / π = 40.7 in / 103 cm diameter.

Border depth and the mini target
The scallops add an unverified amount at the edge, and blocking can move the result within or beyond the source target ranges. Note the mini's unbordered derived diameter (17.8 in / 45 cm) sits just under the bottom of its own 18–21 in / 46–53 cm target, so that target depends on border depth plus blocking; record a blocked sample before repeating the range. Measure the relaxed centre opening, unbordered body and finished blocked skirt. Do not force-block severe cupping or ruffling into a claimed size.

Abbreviation	Meaning	Abbreviation	Meaning
BO	5-dc bobble in US terms / 5-tr bobble in UK terms; counts as 1 stitch	R / rnd	round
CC	contrast colour	sc = dc	US single crochet / UK double crochet
ch	chain	sl st	slip stitch
dc = tr	US double crochet / UK treble crochet	st(s)	stitch(es)
FO	fasten off	(n)	stitch count at round end
MC	main colour		

Every construction round appears twice side by side: US left, UK right, identical counts. Prose uses US terms unless both names are shown.

Construction & Techniques

Joined rounds without turning
The starting ch 2 never counts as a stitch. Work the first dc or BO into the SAME stitch as the join, mark it, and slip stitch to that marked stitch at round end. Do not work into the ch 2 or joining slip stitch.

1. Fit the closed centre first
make the relaxed ch-20 ring and pass it over the actual central pole or trunk area before R1. The written ch-24 option makes a larger ring without changing R1, but its finished opening is still yarn- and tension-dependent. If neither relaxed ring fits without stretching or interfering with the stand, this closed-centre construction is not suitable; do not cut the finished skirt to create a slit.

2. Twelve-repeat circle
Round N consumes N - 1 old stitches and produces N new stitches in each of 12 repeats. The round therefore ends with exactly 12 x N stitches. R9 requires seven plain dc per repeat and R10 requires eight; shortening either repeat leaves stitches unworked.

3. Five-dc bobble yarn over, insert the hook, pull up a loop, yarn over and pull through two loops, leaving the incomplete dc on the hook. Repeat four more times in the same stitch until six loops remain; yarn over and pull through all six. Push the completed one-stitch bobble to the right side.	**4. Original colour route** work R1–R4 and every non-bobble round in forest green MC. Work each complete bobble round — R5, R8, R11 and every third round thereafter — in oat cream CC; the plain dc and increase stitches on that round are CC too. Use CC again for the optional spokes and border. To change colour, work the round-ending slip stitch with the colour required for the next round. Carry a resting colour loosely up the wrong-side join across the intervening round or rounds, or cut and secure it; never strand yarn across the circle or pull the carried section tight.
5. Track the increase columns if adding surface spokes, place one locking marker in the second dc of each R2 increase — R2 has exactly 12 increases, so this gives exactly 12 markers. On every later round, the marked stitch is the last old stitch consumed by its repeat; move that marker to the second dc of the new increase. The 12 final marker positions provide evenly spaced outer endpoints.	**6. Surface slip stitch** hold the yarn behind the work, then insert the hook from the front of the fabric at the next point on the pinned radial guide, pull up a loop, and pull it through the loop already on the hook. The bobbles mark the front; the yarn supply stays behind it for the whole spoke. Keep every slip stitch relaxed and check the fabric after each spoke.

Instructions - Centre Ring & Growth Rounds

Ch 2 starts every round and is never counted. Work the first dc - or first BO on a bobble round - into the same stitch as the join, then end with a slip stitch to that first actual stitch. Legend: grey rows = bobble round in CC, white rows = plain round in MC; the outlined square is your printable progress box.

Centre ring & Rounds 1-8

	Rnd	US terms	UK terms	Sts	Note
	Ring	Ch 20, check the relaxed ring against the stand, then sl st to first ch without twisting	Ch 20, check the relaxed ring against the stand, then sl st to first ch without twisting	–	ch 24 is the only written larger-opening option; recheck before R1
	R1	Ch 2, 12 dc in ring, sl st to first dc	Ch 2, 12 tr in ring, sl st to first tr	**(12)**	–
	R2	Ch 2, 2 dc in each st around, sl st to first dc	Ch 2, 2 tr in each st around, sl st to first tr	**(24)**	12 increases; +12 every round from here
	R3	Ch 2, [dc in next st, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next st, 2 tr in next st] x 12, sl st to first tr	**(36)**	–
	R4	Ch 2, [dc in next 2 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 2 sts, 2 tr in next st] x 12, sl st to first tr	**(48)**	last MC round before the first bobble round
	R5	Ch 2, [**BO in next st**, dc in next 2 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 2 sts, 2 tr in next st] x 12, sl st to first BO	**(60)**	first bobble round · CC
	R6	Ch 2, [dc in next 4 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 4 sts, 2 tr in next st] x 12, sl st to first tr	**(72)**	–
	R7	Ch 2, [dc in next 5 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 5 sts, 2 tr in next st] x 12, sl st to first tr	**(84)**	–
	R8	Ch 2, [**BO in next st**, dc in next 5 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 5 sts, 2 tr in next st] x 12, sl st to first BO	**(96)**	bobble round · CC

Rounds 9-16

	Rnd	US terms	UK terms	Sts	Note
	R9	Ch 2, [dc in next 7 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 7 sts, 2 tr in next st] x 12, sl st to first tr	**(108)**	seven plain dc per repeat
	R10	Ch 2, [dc in next 8 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 8 sts, 2 tr in next st] x 12, sl st to first tr	**(120)**	eight plain dc per repeat
	R11	Ch 2, [**BO in next st**, dc in next 8 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 8 sts, 2 tr in next st] x 12, sl st to first BO	**(132)**	bobble round · CC
	R12	Ch 2, [dc in next 10 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 10 sts, 2 tr in next st] x 12, sl st to first tr	**(144)**	10 plain + 1 increase anchor consumes 11 and makes 12
	R13	Ch 2, [dc in next 11 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 11 sts, 2 tr in next st] x 12, sl st to first tr	**(156)**	–
	R14	Ch 2, [**BO in next st**, dc in next 11 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 11 sts, 2 tr in next st] x 12, sl st to first BO	**(168)**	**MINI SIZE — stop growth; optional spokes, then border**
	R15	Ch 2, [dc in next 13 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 13 sts, 2 tr in next st] x 12, sl st to first tr	**(180)**	–
	R16	Ch 2, [dc in next 14 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 14 sts, 2 tr in next st] x 12, sl st to first tr	**(192)**	–

Count check for the R11-to-R12 step
R11 consumes 10 old stitches per repeat (1 BO + 8 plain dc + 1 increase anchor) and produces 11, giving 132. R12 consumes 11 old stitches per repeat (10 plain dc + 1 increase anchor) and produces 12, giving 144. R12 therefore requires 10 plain dc, not 9; using 9 would leave 12 old stitches unworked and produce only 132.

Rounds 17-24

	Rnd	US terms	UK terms	Sts	Note
	R17	Ch 2, [**BO in next st**, dc in next 14 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 14 sts, 2 tr in next st] x 12, sl st to first BO	**(204)**	bobble round · CC
	R18	Ch 2, [dc in next 16 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 16 sts, 2 tr in next st] x 12, sl st to first tr	**(216)**	–
	R19	Ch 2, [dc in next 17 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 17 sts, 2 tr in next st] x 12, sl st to first tr	**(228)**	–
	R20	Ch 2, [**BO in next st**, dc in next 17 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 17 sts, 2 tr in next st] x 12, sl st to first BO	**(240)**	bobble round · CC
	R21	Ch 2, [dc in next 19 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 19 sts, 2 tr in next st] x 12, sl st to first tr	**(252)**	–
	R22	Ch 2, [dc in next 20 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 20 sts, 2 tr in next st] x 12, sl st to first tr	**(264)**	–
	R23	Ch 2, [**BO in next st**, dc in next 20 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 20 sts, 2 tr in next st] x 12, sl st to first BO	**(276)**	**STANDARD SIZE — stop growth; optional spokes, then border · CC**
	R24	Ch 2, [dc in next 22 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 22 sts, 2 tr in next st] x 12, sl st to first tr	**(288)**	–

Rounds 25-32

	Rnd	US terms	UK terms	Sts	Note
	R25	Ch 2, [dc in next 23 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 23 sts, 2 tr in next st] x 12, sl st to first tr	**(300)**	–
	R26	Ch 2, [**BO in next st**, dc in next 23 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 23 sts, 2 tr in next st] x 12, sl st to first BO	**(312)**	bobble round · CC
	R27	Ch 2, [dc in next 25 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 25 sts, 2 tr in next st] x 12, sl st to first tr	**(324)**	–
	R28	Ch 2, [dc in next 26 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 26 sts, 2 tr in next st] x 12, sl st to first tr	**(336)**	–
	R29	Ch 2, [**BO in next st**, dc in next 26 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 26 sts, 2 tr in next st] x 12, sl st to first BO	**(348)**	bobble round · CC
	R30	Ch 2, [dc in next 28 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 28 sts, 2 tr in next st] x 12, sl st to first tr	**(360)**	–
	R31	Ch 2, [dc in next 29 sts, 2 dc in next st] x 12, sl st to first dc	Ch 2, [tr in next 29 sts, 2 tr in next st] x 12, sl st to first tr	**(372)**	–
	R32	Ch 2, [**BO in next st**, dc in next 29 sts, 2 dc in next st] x 12, sl st to first BO	Ch 2, [BO in next st, tr in next 29 sts, 2 tr in next st] x 12, sl st to first BO	**(384)**	**LARGE SIZE — stop growth; optional spokes, then border · CC**

Spokes, Scalloped Border & Sizes

Optional surface snowflake spokes - work before the border. Leave the 12 outer increase-column markers in place; pin a straight radial guide from the foundation ring to each marker. Use a separate CC length per spoke; do not carry yarn from one outer edge back to the centre. Surface stitches decorate the fabric and do not add to the growth-round counts.

	Rnd	US terms	UK terms	Sts	Note
	Spoke	Join CC through the fabric immediately outside the centre ring; work relaxed surface sl sts outward along one pinned guide to its marked increase column; FO at the outer edge	Join CC through the fabric immediately outside the centre ring; work relaxed surface sl sts outward along one pinned guide to its marked increase column; FO at the outer edge	–	1 separate spoke; no reverse-side float
	Repeat	Repeat the complete join-to-FO spoke route separately at each remaining marked column	Repeat the complete join-to-FO spoke route separately at each remaining marked column	–	12 spokes total; count 12 centre starts and 12 outer ends

Scalloped border - work into the final growth round only after the size and spokes are approved.

	Rnd	US terms	UK terms	Sts	Note
	Border	Join CC in any st; ch 1, [sc in next st, skip 2 sts, 5 dc in next st, skip 2 sts] around, sl st to first sc, FO	Join CC in any st; ch 1, [dc in next st, skip 2 sts, 5 tr in next st, skip 2 sts] around, sl st to first dc, FO	same worked-st total as the final growth round	each repeat consumes 6 edge sts and makes 1 scallop

Why the border closes exactly

Each repeat consumes six final-round anchors and produces six worked stitches (one sc plus a five-dc shell): 168 / 6 = 28 scallops, 276 / 6 = 46, 384 / 6 = 64. The join stitch is taken by the final skip of the last repeat, so the closing slip stitch lands one stitch past the join - correct, and required for exact closure. Never improvise a partial shell.

Size	Stop after round	Stitches	Scallops	Unverified finished target
Mini / tabletop	R14	168	28	18–21 in / 46–53 cm
Standard	R23	276	46	29–33 in / 74–84 cm
Large	R32	384	64	38–43 in / 97–109 cm

Finishing, Assembly & Troubleshooting

1. Confirm the selected final growth round contains 168, 276 or 384 stitches before adding surface work.
2. Work the optional 12 separate spokes, checking flatness after each, then work the exact scalloped border.
3. Weave every spoke, colour-change and border tail on the wrong side in at least two directions without tightening the radius.
4. Block only by a method suitable for the actual fibre. Pin to a flat circle without stretching the centre opening; never place an iron directly on the yarn, and do not apply heat unless the yarn maker permits the chosen method.
5. When fully dry and relaxed, record the centre opening, diameter through several axes, mass and edge condition. Confirm the opening fits the intended stand without affecting its stability, reservoir or fittings.
6. Install the closed ring before final tree assembly and only in a way permitted by the tree and stand manufacturers. Reinspect the complete arrangement after lights and cords are positioned.

Troubleshooting - problem / fix

The relaxed centre ring does not fit	stop before R1. Try the written ch-24 option and recheck it. Do not stretch the ring over hardware, cut the finished skirt or assume blocking will create a safe fit.
The body ruffles	first recount the complete round and confirm there are exactly 12 increases. If the count is correct, let the gauge piece rest and check both gauge directions; test another hook or yarn before continuing. Do not randomly omit increases because that breaks the written size and border route.
The body cups	check for a short repeat or skipped increase, especially R9 and R10. If the count is correct, expect the mild cupping tendency described in Gauge & size and reassess hook, yarn and the stitch gauge rather than adding unrecorded stitches.
A surface spoke puckers the skirt	remove that spoke and remake it with looser slip stitches. Each spoke needs its own centre join and outer-edge fasten-off; there should be no long float on the wrong side.
The border does not close	verify the final total is 168, 276 or 384 and consume exactly six anchors per scallop. Do not improvise a partial shell.
Bobbles stay flat	keep the five incomplete dc relaxed, close all six loops together and push the bobble to the right side before the next stitch.

Colourways & Care

Colour	Role	Alternative palettes
FOREST GREEN (MC)	every non-bobble growth round	snow white MC + deep red CC
OAT CREAM (CC)	bobble rounds, spokes + border	navy MC + silver-grey CC; or single-colour cream

Use comparable worsted/aran yarns so colour changes do not alter gauge. Metallic filament, sequins, beads and battery-light additions are not part of this pattern and need separate handling, care and safety review.

Care & storage

Follow the most restrictive instruction among every yarn and added material. Before publishing a cleaning claim or supplying finished skirts, clean and fully dry a complete full-size sample by the proposed method. Remeasure the diameter and opening and inspect dye transfer, shrinkage, bobble shape, surface-spoke tension, joins and scallops.

Until that complete-sample test passes, do not make a machine-wash or tumble-dry claim. Recommend gentle surface cleaning only. Store completely dry and away from heat, flame, damp and pests; roll or lay flat rather than sharply creasing the bobble field. Inspect again before seasonal use.

Terms, Licence & Finish Checklist

Terms of Use

Personal & Small-Batch License: This pattern is licensed to the purchaser for personal use and for the small-batch sale of finished physical items. Listings for finished items must credit “Pattern by Novality Store · Design Code NS-14”.

Restrictions: You may not copy, share, translate, upload, redistribute, resell, or publish the pattern or any substantial part of it in digital or printed form.

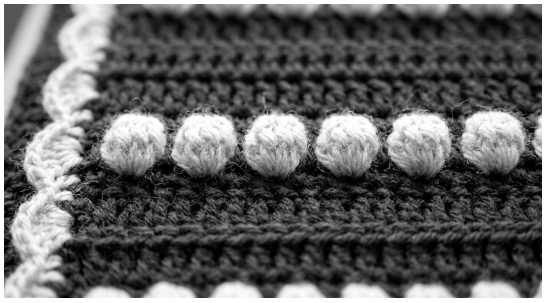
Seller responsibilities: Finished-item listings must not claim unverified safety, testing, size, yarn quantity, fit or care results. The maker or seller remains responsible for materials, intended use, classification, assessment, testing, warnings, documentation, labelling and traceability wherever a finished item is supplied.

- ☐ Centre ring fits the stand relaxed; join and ends secured
- ☐ Every growth round counted; 12 increase columns aligned
- ☐ Final count confirmed: 168 / 276 / 384
- ☐ Optional spokes: 12 separate routes, no puckering or floats
- ☐ Border closed: 28 / 46 / 64 complete scallops
- ☐ Blocked dimensions recorded; stand and cord clearances checked

Project notes

Finished-item record

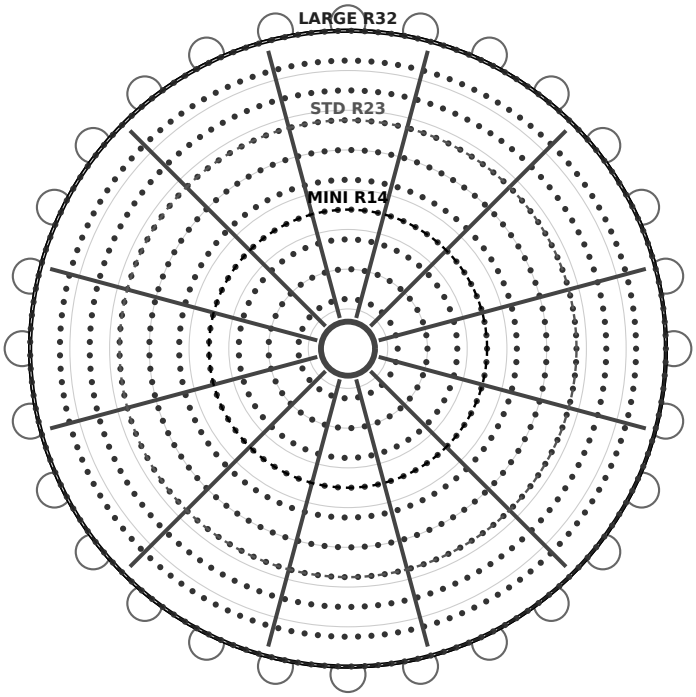
Yarn brand, fibre & dye lot - MC	
Yarn brand, fibre & dye lot - CC	
Hook used & gauge achieved (sts / rows per 10 cm)	
Blocked diameter, stand model & cord clearance check	
Finish date & maker	



Stitch detail: oat cream five-dc bobbles and the scalloped shell edge over forest green dc. Artwork is illustrative of the written design.

Keep this record with your yarn labels: dye lots and gauge are the two values buyers and future repairs most often need.

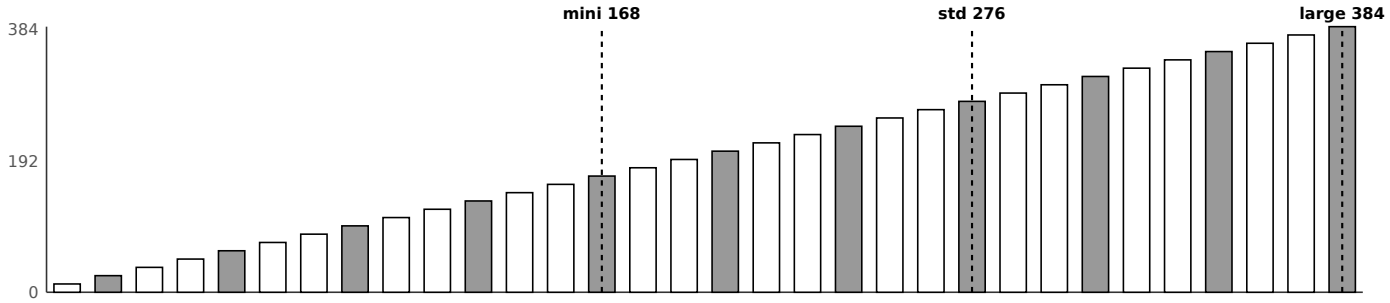
Appendix A - Visual Schematics, Assembly Map & Growth Chart



Top-view schematic (not to scale): closed centre ring, 12 surface-crochet spokes, bobble rings every third round from R5 (black dots), faint growth rings, scalloped shell edge, and the three dashed size stops.

1 Fit centre ring	2 Growth rounds	3 Surface spokes	4 Scalloped border	5 Finish
ch-20 ring over the stand before R1 - recheck ch-24	R1-R32, +12 sts per round stop at R14 / R23 / R32	12 separate CC routes, centre ring to outer edge	6 anchors per scallop: 28 / 46 / 64, then FO	weave tails, block flat, record dimensions, install

Order matters: spokes are worked before the border; the border is worked into the final growth round only.



Stitches per round, R1-R32. Grey bars = bobble rounds (every third round from R5). Dashed lines = the three size stops.

Appendix B - Machine-Verification & 5-Axiom Audit

Every round was re-derived from the written repeat and checked against the pattern's own rule: round N consumes N - 1 old stitches and produces N new stitches in each of 12 repeats, ending at exactly 12 x N, so no stitch is left unworked. All 32 rounds pass.

Rnd	Type	Cons.	Prod.	x 12	Stated	OK
1	plain	0	1	12	12	**OK**
2	plain	1	2	24	24	**OK**
3	plain	2	3	36	36	**OK**
4	plain	3	4	48	48	**OK**
5	bobble	4	5	60	60	**OK**
6	plain	5	6	72	72	**OK**
7	plain	6	7	84	84	**OK**
8	bobble	7	8	96	96	**OK**
9	plain	8	9	108	108	**OK**
10	plain	9	10	120	120	**OK**
11	bobble	10	11	132	132	**OK**
12	plain	11	12	144	144	**OK**
13	plain	12	13	156	156	**OK**
14	bobble	13	14	168	168	**OK**
15	plain	14	15	180	180	**OK**
16	plain	15	16	192	192	**OK**

Rnd	Type	Cons.	Prod.	x 12	Stated	OK
17	bobble	16	17	204	204	**OK**
18	plain	17	18	216	216	**OK**
19	plain	18	19	228	228	**OK**
20	bobble	19	20	240	240	**OK**
21	plain	20	21	252	252	**OK**
22	plain	21	22	264	264	**OK**
23	bobble	22	23	276	276	**OK**
24	plain	23	24	288	288	**OK**
25	plain	24	25	300	300	**OK**
26	bobble	25	26	312	312	**OK**
27	plain	26	27	324	324	**OK**
28	plain	27	28	336	336	**OK**
29	bobble	28	29	348	348	**OK**
30	plain	29	30	360	360	**OK**
31	plain	30	31	372	372	**OK**
32	bobble	31	32	384	384	**OK**

Cons. = stitches consumed per repeat (N - 1); Prod. = new stitches per repeat (N); x 12 = the twelve repeats; Stated = the pattern's printed count; OK = consumed x 12 equals the previous stated total, so no stitch is left unworked. Round 1 consumes only the ch-20 joining ring. Border closures: mini 168 / 6 = 28 scallops, standard 276 / 6 = 46, large 384 / 6 = 64. Negative control: the same ladder at 27 repeats fails every closure, confirming the check discriminates.

Verification summary

32 / 32 growth rounds pass the consumption and production rule with zero stitches left unworked; bobble cadence R5-R32 every third round confirmed; border closure exact at 28 / 46 / 64 with a 27-repeat negative control failing as expected; all 36 US/UK row pairs verified as exact dialect translations; hook conversion 5 mm = US H/8 and 5.5 mm = US I/9 confirmed against Craft Yarn Council tables. Verification Method: Audited via deterministic static analysis (crochet-check v1.0.0 engine) and 5-Axiom mathematical proofs. All 32 growth rounds and border closures pass 100%.

5-Axiom audit summary (formal)

Axiom	What it guarantees	Evidence in this pattern	Result
1 · Domain Isolation	Rounds never count another piece's stitches	Spokes and border excluded from every growth sum	**PASS**
2 · Edge Lineage	Consumed counts trace to earlier stated totals	Round N consumes N - 1 per repeat, exactly as stated	**PASS**
3 · Open Boundaries	No seam aimed at a closed ring or cinched point	Live open edge: 6 anchors per scallop, 28 / 46 / 64	**PASS**
4 · 2-Manifold Conservation	Every edge worked exactly once	32 / 32 rounds: consumed N - 1, produced N, none left	**PASS**
5 · Gauge Bounding	Gauge inside the published worsted band	5-5.5 mm hook; size stops R14 / R23 / R32 bounded	**PASS**

Audit executed by the deterministic crochet-check v1.0.0 static-analysis engine plus the count ladder above; no sampled or hand-checked results. Engine stages live: yes.

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